

Linux Task 03: Process Management, System Monitoring & Basic Shell Scripting

Screenshots

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nimitha@LAPTOP: ~
nimitha@LAPTOP:~/Linux_Task_03_Nimitha$ ps
  PID TTY          TIME CMD
  343 pts/0        00:00:00 bash
 1808 pts/0        00:00:00 ps
nimitha@LAPTOP:~/Linux_Task_03_Nimitha$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  24576 15220 ?        Ss   12:03   0:05 /sbin/init
root           2  0.0  0.0   3172  2208 ?        S<   12:03   0:00 [init]
root           7  0.0  0.0   3172  2040 ?        S<   12:03   0:00 [plan9 --c]
root          53  0.0  0.2  50380 16520 ?        S<S  12:03   0:03 /usr/lib/
systemd+      87  0.0  0.1  22528 14124 ?        Ss   12:03   0:01 /usr/lib/
root         102  0.2  0.1  35160 11916 ?        Ss   12:03   0:23 /usr/lib/
root         154  0.0  0.0   2888  1912 ?        Ss   12:03   0:00 /bin/sh /
root         155  0.0  0.0   4464  2980 ?        Ss   12:03   0:00 /usr/sbin
message+     156  0.0  0.0   8796  5196 ?        Ss   12:03   0:01 @dbus-dae
root         162  0.0  0.3  44080 29496 ?        Ss   12:03   0:00 /usr/bin/
redis        163  0.6  0.1  80228 15156 ?        Ssl  12:03   1:08 /usr/bin/
root         171  0.0  0.1  18336  9252 ?        Ss   12:03   0:00 /usr/lib/
syslog       179  0.0  0.0  220532 56444 ?        Ssl  12:03   0:01 /usr/sbin
root         216  0.0  0.0   5484  2700 tty1     Ss+  12:03   0:00 /usr/sbin
_chrony      249  0.0  0.0   40080  7580 ?        Ss   12:03   0:00 /usr/sbin
_chrony      264  0.0  0.0   12080  2404 ?        S   12:03   0:00 /usr/sbin
root         300  0.0  0.4 123444  32240 ?        Ss   12:03   0:00 /usr/bin/
root         341  0.0  0.0   3176  1112 ?        Ss   12:03   0:00 [init]
root         342  0.0  0.0   3192  1248 ?        Ss   12:03   0:00 [init]
nimitha      343  0.0  0.0   6260  5484 pts/0    Ss   12:03   0:00 bash
root         344  0.0  0.0   8628  5408 ?        Ss   12:03   0:00 login --
nimitha      422  0.0  0.1  22736 13240 ?        Ss   12:03   0:00 /usr/lib/
nimitha      424  0.0  0.0   22896  4044 ?        Ss   12:03   0:00 (sd-pam)
nimitha      450  0.0  0.0   6128  5356 pts/1    Ss+  12:03   0:00 bash
root        1650  0.0  0.1 1866280 15152 ?        Ssl  14:39   0:00 /usr/libe
nimitha     1818  0.0  0.0   7152  4228 pts/0    R+   14:50   0:00 ps aux
nimitha@LAPTOP:~/Linux_Task_03_Nimitha$ top
top - 14:50:20 up 2:46, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7560.1 total, 6986.9 free, 530.9 used, 191.8 buff/cache
MiB Swap: 2048.0 total, 2088.0 free, 0.0 used, 7029.2 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM     TIME+ COMMAND
  163 redis      20   0   80228  15156 11616 S   0.7   0.20   1:09.08 /usr/bin/
```

```
nimitha@LAPTOP: ~  
nimitha@LAPTOP:~$ ps aux | grep sleep  
nimitha 2123 0.0 0.0 16104 7584 pts/0 S+ 16:11 0:00 sleep 300  
nimitha 2197 0.0 0.0 4128 2248 pts/2 S+ 16:12 0:00 grep --color=auto  
nimitha@LAPTOP:~$ kill 2123
```

```
0[ 0.0%] 4[ 0.7%]  
1[ 0.0%] 5[ 0.0%]  
2[ 0.7%] 6[ 0.0%]  
3[ 0.0%] 7[ 0.0%]  
Mem[||||] 411M/7.38G Tasks: 27, 19 thr, 0 kthr; 2 running  
Swp[ ] 0K/2.00G Load average: 0.18 0.05 0.01  
Uptime: 02:47:56
```

Main		I/O											
PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%	TIME+	Command		
163	redis	20	0	80228	15156	11616	S	0.7	0.2	1:09.37	/usr/bin/		
1	root	20	0	24576	15224	11228	S	0.0	0.2	0:05.18	/sbin/init		
2	root	20	0	3172	2208	2072	S	0.0	0.0	0:00.06	/init		
7	root	20	0	3172	2040	1940	S	0.0	0.0	0:00.00	plan9 --c		
8	root	20	0	3172	2040	0	S	0.0	0.0	0:00.00	plan9 --c		
9	root	20	0	3172	2208	0	S	0.0	0.0	0:00.00	/init		
53	root	19	-1	50380	16536	15232	S	0.0	0.2	0:03.24	/usr/lib/		
87	systemd-re	20	0	22528	14124	11516	S	0.0	0.2	0:01.57	/usr/lib/		
102	root	20	0	35160	11916	8880	S	0.0	0.2	0:23.10	/usr/lib/		
154	root	20	0	2888	1912	1792	S	0.0	0.0	0:00.14	/bin/sh /		
155	root	20	0	4464	2980	2732	S	0.0	0.0	0:00.27	/usr/sbin		
156	messagebus	20	0	8796	5240	4544	S	0.0	0.1	0:01.42	@dbus-dae		
162	root	20	0	44080	29496	14364	S	0.0	0.4	0:00.70	/usr/bin/		
171	root	20	0	18336	9252	7980	S	0.0	0.1	0:00.85	/usr/lib/		
179	syslog	20	0	215M	5644	4360	S	0.0	0.1	0:00.19	/usr/sbin		
205	syslog	20	0	215M	5644	0	S	0.0	0.1	0:00.30	/usr/sbin		
206	syslog	20	0	215M	5644	0	S	0.0	0.1	0:00.79	/usr/sbin		
207	syslog	20	0	215M	5644	0	S	0.0	0.1	0:00.15	/usr/sbin		
216	root	20	0	5484	2700	2512	S	0.0	0.0	0:00.01	/usr/sbin		
249	_chrony	20	0	20408	7580	6776	S	0.0	0.1	0:00.67	/usr/sbin		
264	_chrony	20	0	12080	2404	1764	S	0.0	0.0	0:00.04	/usr/sbin		
300	root	20	0	120M	32240	17644	S	0.0	0.4	0:00.50	/usr/bin/		
306	redis	20	0	80228	15156	0	S	0.0	0.2	0:00.00	/usr/bin/		
307	redis	20	0	80228	15156	0	S	0.0	0.2	0:00.00	/usr/bin/		
308	redis	20	0	80228	15156	0	S	0.0	0.2	0:00.00	/usr/bin/		
309	redis	20	0	80228	15156	0	S	0.0	0.2	0:00.00	/usr/bin/		
310	redis	20	0	80228	15156	0	S	0.0	0.2	0:00.06	/usr/bin/		
317	root	20	0	120M	32240	0	S	0.0	0.4	0:00.00	/usr/bin/		
341	root	20	0	3176	1112	980	S	0.0	0.1	0:00.00	/init		

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice -F8Nice +F9Kill F10Quit

```
nimitha@LAPTOP: ~  
nimitha@LAPTOP:~/Linux_Task_03_Nimitha$ who  
nimitha pts/0 2026-06-14 16: Nimitha$  
nimitha@LAPTOP:~/Linux_Task_03_Nimitha$ w  
16:54:32 up 3:44, 1 user, load average: 0.01, 0.01, 0.00  
USER=====
```

System Memory:						
System	total	used	used	free	buff/cache	avail
Mem:	7.4Gi	561Mi	6.8Gi	3.7Mi	224Mi	6
Swap:	2.0Gi	0B	2.0Gi			

```
Disk Usage:  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/sdd        1007G  2.1G  954G   1% /  
  
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```

Report generated successfully.

```
nimitha@LAPTOP:~/Linux_Task_03_Nimitha$ |
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